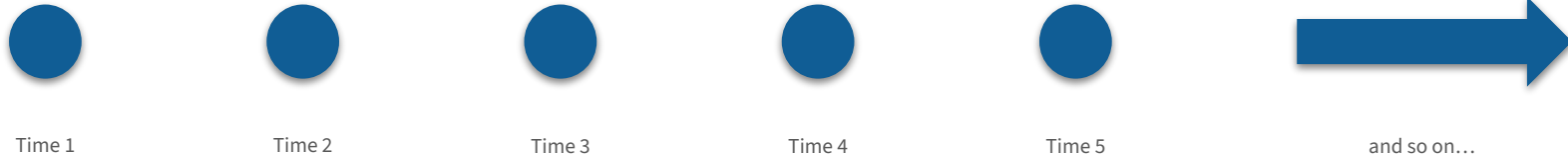


What is a Sequence?

Transformers – Conceptual Foundations

In the context of data science, sequences are any data that observe a chronological order.



Sequence are everywhere!

- Time series data, like stock market prices. 
- Speech, which is a sequence of phonemes. 
- Music, a sequence of notes. 
- Text, a sequence of words. 
- Video, which can be seen as a sequence of image frames. 



Individual units interact to create meaning.

The dog jumped over the boy, and it fell on the grass.



Individual units interact to create meaning.

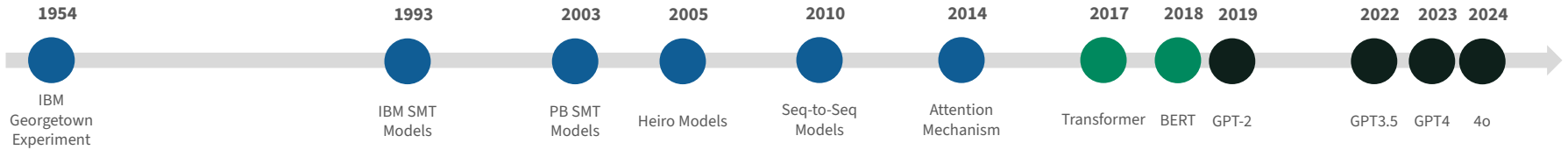
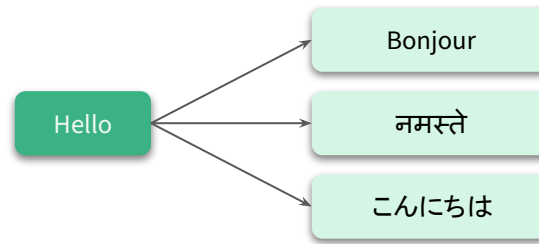
The dog jumped over the boy, and it fell on the grass.

History of Machine Translation

Transformers – Conceptual Foundations



Convert one language to another



What are Seq2Seq Models?

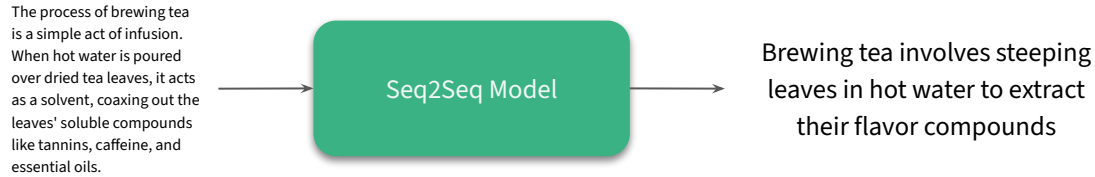
Transformers – Conceptual Foundations



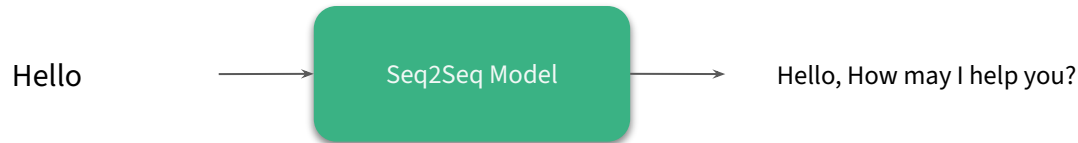
Machine Translation

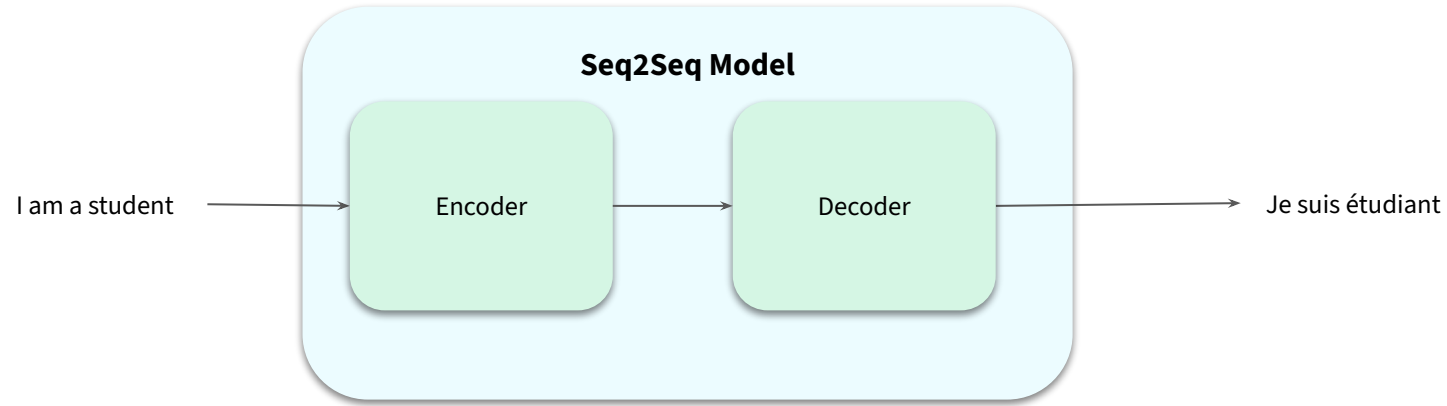


Summarization



Chatbots

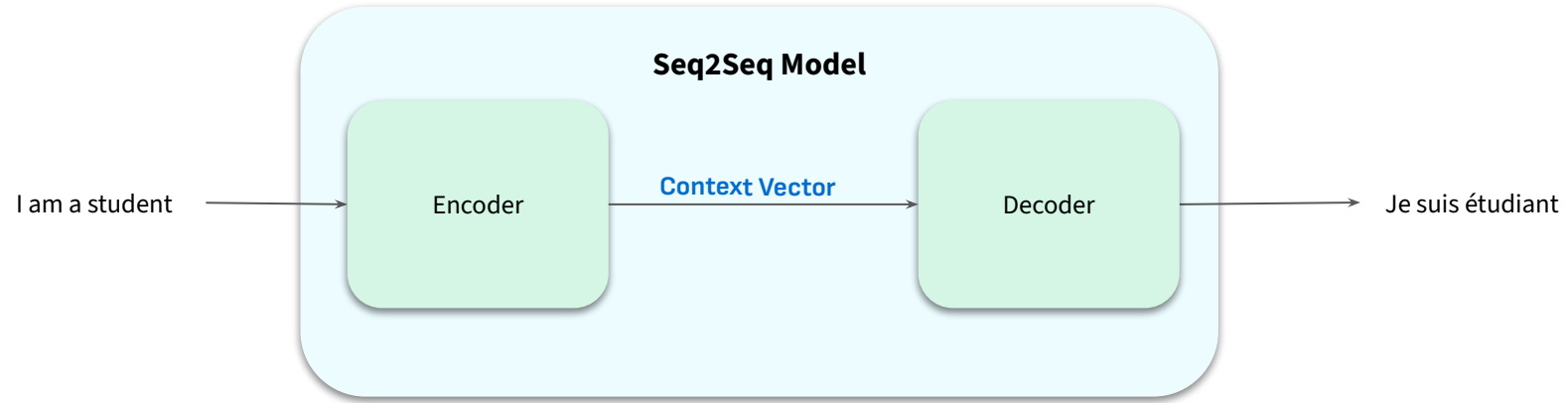






Tangent



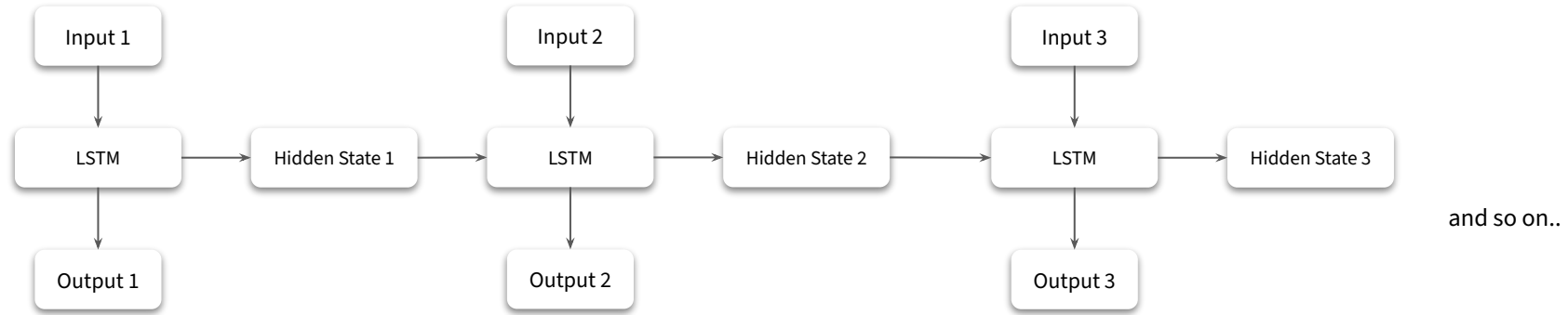


What are RNN-based Seq2Seq Models?

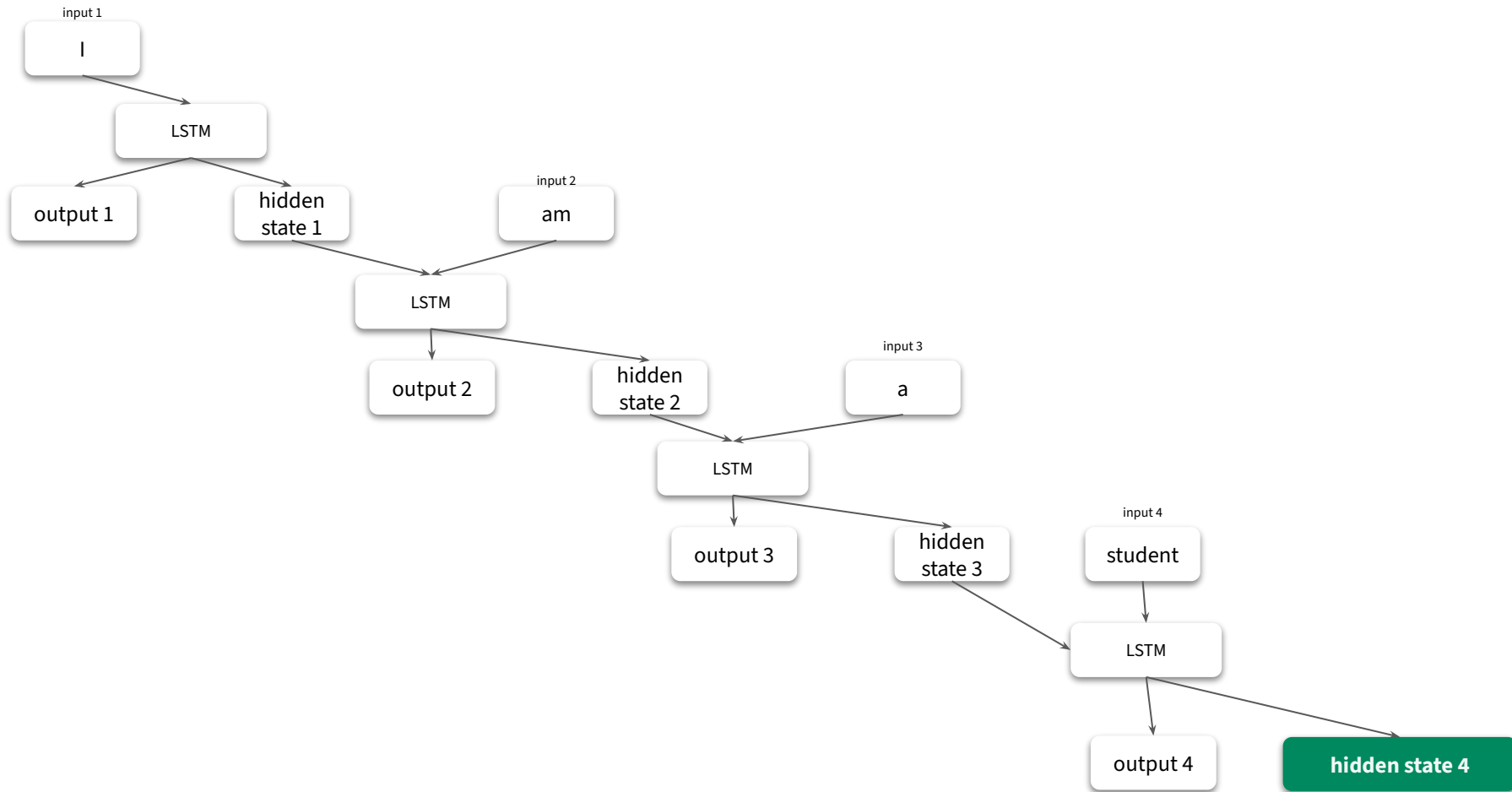
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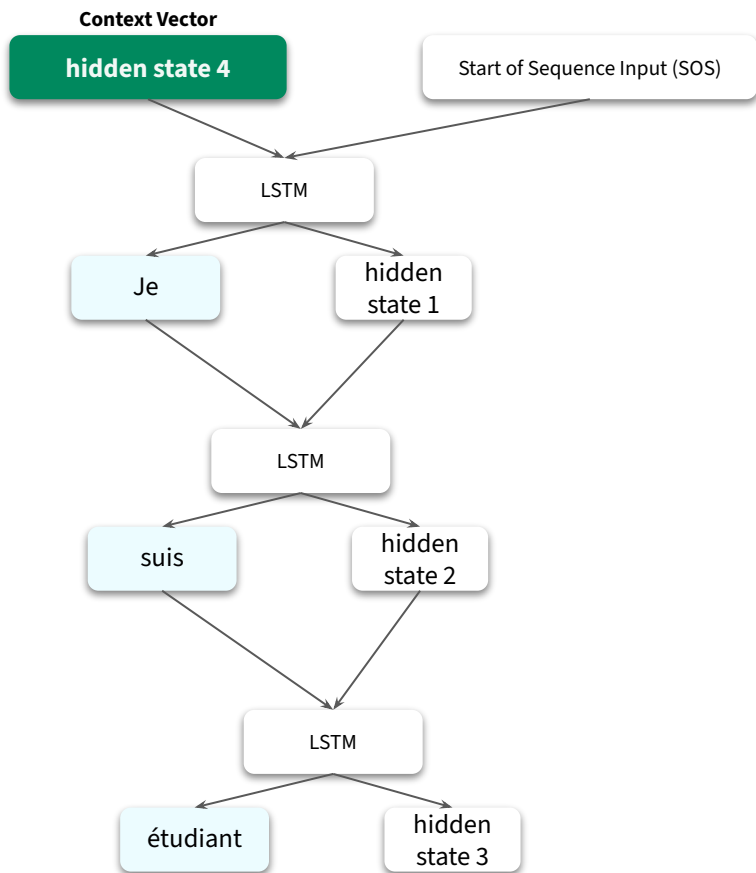
Recurrent Neural Networks



Encoder RNN



Decoder RNN



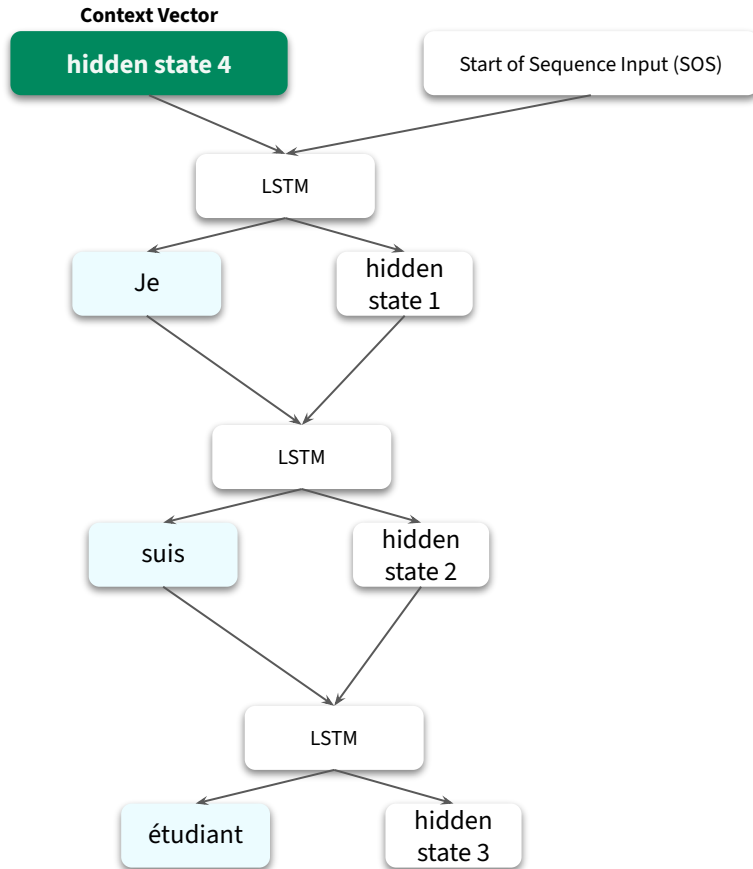


Will this perform well with long sequences? **No.**

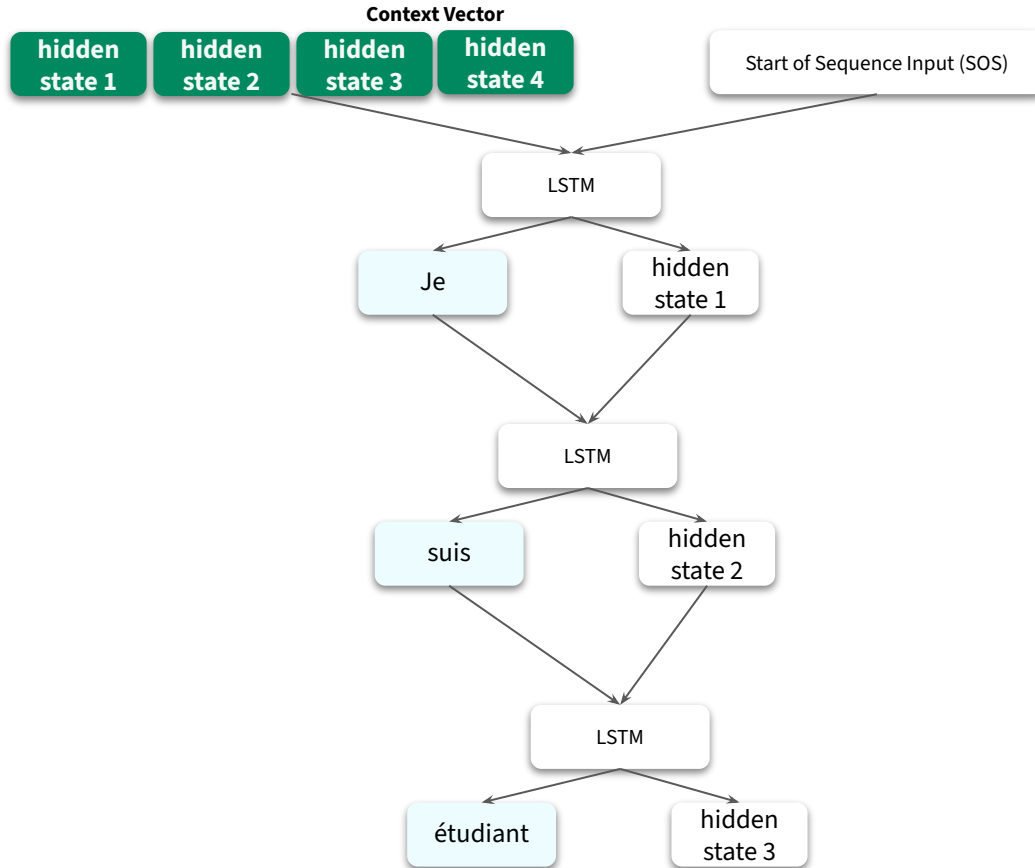
What is Attention?

Transformers – Conceptual Foundations

Decoder RNN with Attention



Decoder RNN with Attention



Interpretability with Attention

	I	am	a	student
Je	Encoder hidden state 1	Encoder hidden state 1	Encoder hidden state 1	Encoder hidden state 1
suis	Encoder hidden state 2	Encoder hidden state 2	Encoder hidden state 2	Encoder hidden state 2
étudiant	Encoder hidden state 3	Encoder hidden state 3	Encoder hidden state 3	Encoder hidden state 3



Attention != Transformer

Attention Model Mechanism.

Your First High-Level Look at Transformers

Transformers – Conceptual Foundations



Attention is all you need!

Transformers uses RNNs?



Attention is all you need!

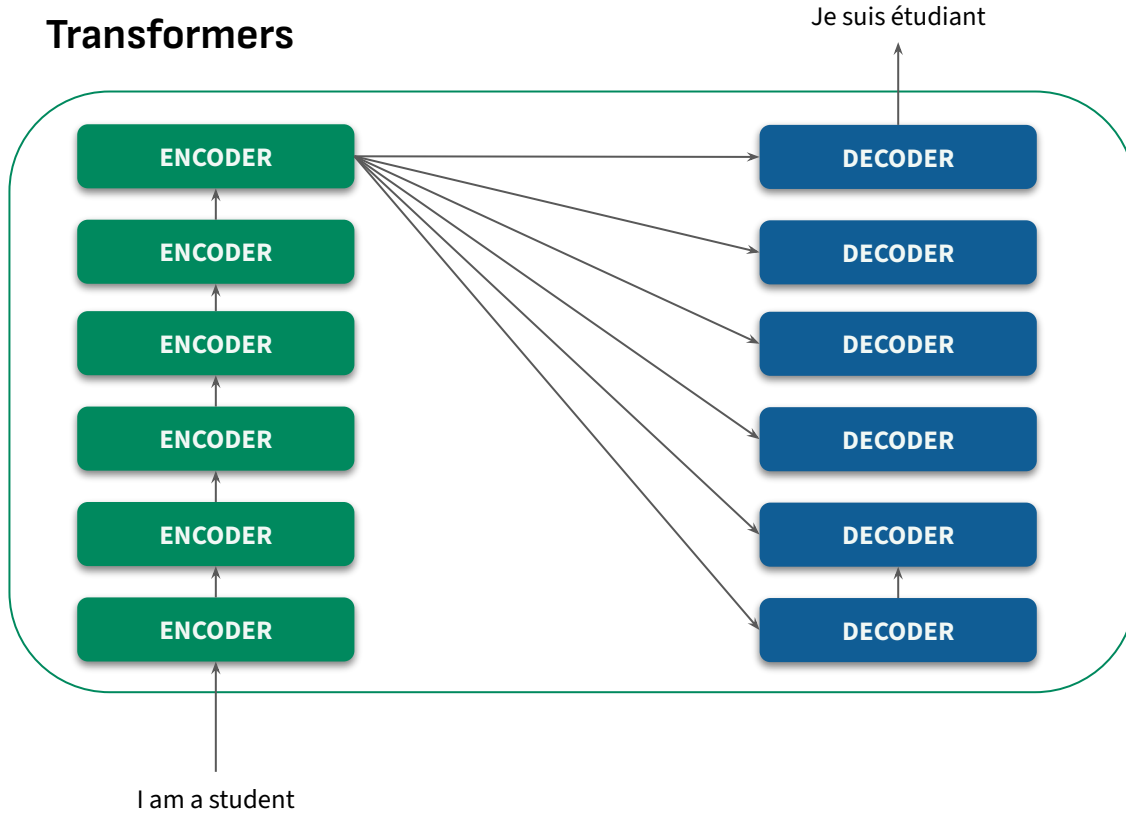
Transformers uses ~~RNNs~~ Attention?



Attention is all you need!

Transformers uses RNNs Attention Self-Attention

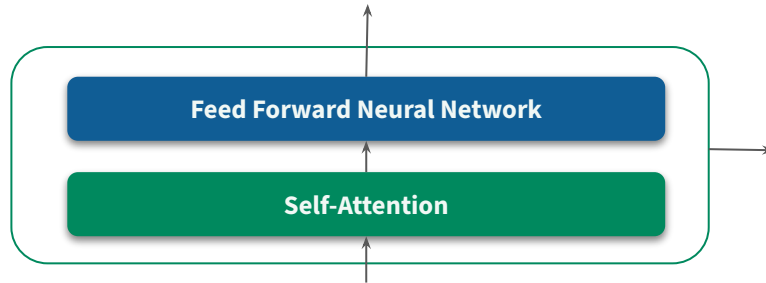
Transformers



The Encoding Component

Transformers – Conceptual Foundations

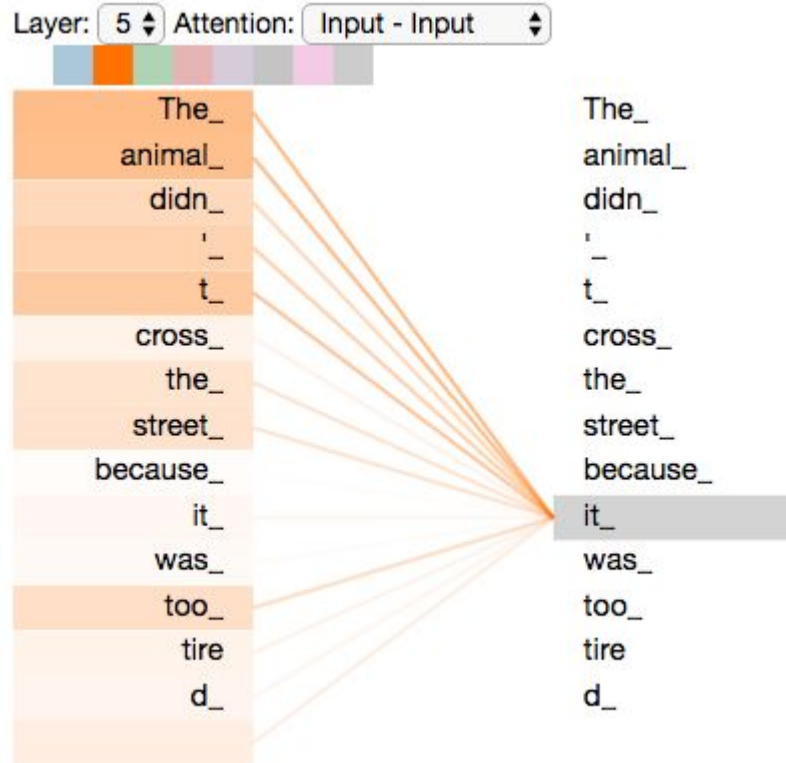
Encoders



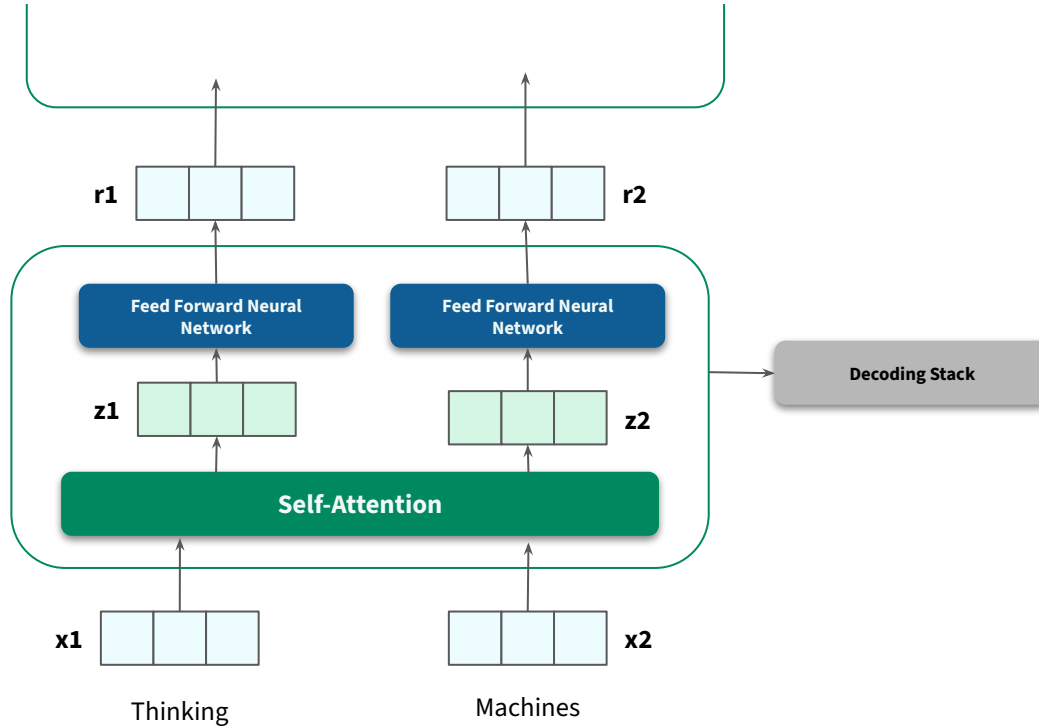
Self-Attention

Transformers – Conceptual Foundations

Self-Attention



Encoders





Questions

Why is the data going to one single self-attention layer rather than two separate self-attention layers?

Why have the feed forward network layers, at all?

How are attention scores calculated?

How Attention Scores are Calculated?

Transformers – Conceptual Foundations



The Red Fox

The red fox, a creature of cunning and adaptability, is a common sight in both rural and urban environments. Despite its name, this mammal can be found in a variety of colors, including silver, black, and other variations.

Value

Key

Besides red, what is one other colour a red fox can be?

Query



The Red Fox

The red fox, a creature of cunning and adaptability, is a common sight in both rural and urban environments. Despite its name, this mammal can be found in a variety of colors, including silver, black, and other variations. The most iconic morph, however, presents a fiery red coat that contrasts sharply with its white underbelly, black-tipped ears, and dark legs. Its long, bushy tail, often tipped with a distinctive white blaze, serves multiple purposes: as a warm covering in cold weather, a signal flag to other foxes, and a counterbalance for agile maneuvering.

Scientifically known as *Vulpes vulpes*, the red fox boasts the widest geographical distribution of any member of the order Carnivora. Its range extends across the entire Northern Hemisphere, from the Arctic Circle to North Africa, North America, and Eurasia. This remarkable success is a testament to its incredible ability to thrive in a vast array of habitats, from dense forests and open grasslands to harsh deserts and even the heart of bustling cities. In recent decades, red foxes have become increasingly common in suburban and urban areas, where they have proven adept at navigating human-dominated landscapes and exploiting new food sources.

The diet of the red fox is as varied as its habitat. An opportunistic omnivore, it preys primarily on small rodents like mice and voles, as well as rabbits, squirrels, and birds. However, it will readily supplement its diet with fruits, berries, insects, and carrion. In urban settings, foxes are notorious for scavenging from garbage cans and compost heaps, a behavior that highlights their resourcefulness. Their hunting techniques are a marvel of instinct and intelligence. With keen hearing, they can locate prey moving underground and will often perform a characteristic high pounce, leaping into the air to surprise and pin down their meal.

Socially, red foxes are generally solitary hunters, though they live in small family groups in underground dens, often referred to as "earth." These dens can be complex, with multiple entrances and chambers, and are sometimes inherited and expanded over generations. A dominant pair, the dog fox and the vixen, will mate for life and raise a litter of kits each spring. The kits, born blind and deaf, are cared for by both parents and sometimes by older siblings from a previous litter who act as "helpers." By autumn, the young foxes will have dispersed to establish their own territories.

The relationship between humans and red foxes is complex and multifaceted. In many cultures, the fox is a symbol of cunning and trickery, a prominent figure in folklore and fables. While they can be a nuisance to farmers due to occasional predation on poultry, they also play a crucial role in controlling populations of agricultural pests. Their growing presence in cities has led to a mixture of delight and concern, with many people enjoying the sight of these wild creatures while others worry about potential conflicts. Regardless of human perception, the red fox continues to demonstrate a remarkable resilience, a testament to its evolutionary prowess and its ability to adapt to a changing world.

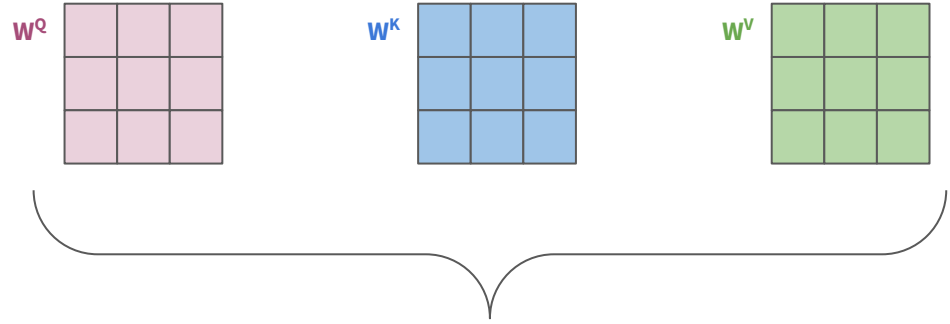


Self-Attention is all you need!

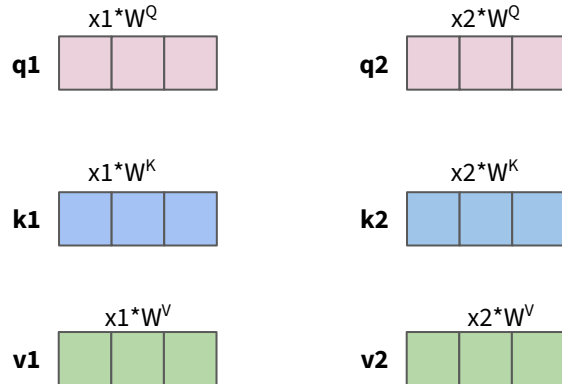
How does Self-Attention understand every word's interaction with every other word in a sequence?



Attention Scores: Setup

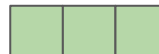


Randomly initialized. Optimized during training.



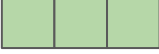
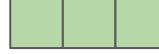
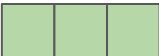
Attention Scores: Softmax Scores



Input	Thinking	Machines
Embeddings	x1 	x2 
Query	$x1 * W^Q$ q1 	$x2 * W^Q$ q2 
Key	$x1 * W^K$ k1 	$x2 * W^K$ k2 
Value	$x1 * W^V$ v1 	$x2 * W^V$ v2 
Score	q1 . k1 = 112	q1 . k2 = 96
Divide by 8	112/8 = 14	96/8 = 12
Softmax Score	0.88	0.12

Attention Scores: Softmax Scores



Input	Thinking	Machines
Embeddings	x1 	x2 
Query	$x1 * W^Q$ q1 	$x2 * W^Q$ q2 
Key	$x1 * W^K$ k1 	$x2 * W^K$ k2 
Value	$x1 * W^V$ v1 	$x2 * W^V$ v2 
Score	q1 . k1 = 112	q1 . k2 = 96
Divide by 8	112/8 = 14	96/8 = 12
Softmax Score	0.88	0.12
Softmax Score * Value	s1 	s2 
Attention Score	z1 	

Multi-headed Self-Attention

Transformers – Conceptual Foundations



Each Head = individual, independent self-attention execution.

Each Head = different way or "lens" to view the same input sequence.



z^1_1			
z^2_1			
z^3_1			
z^4_1			
z^5_1			
z^6_1			
z^7_1			
z^8_1			



w^0



z_1

--	--	--



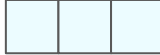
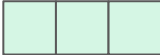
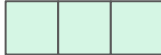
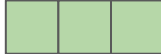
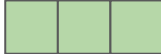
Tangent

Multi-Headed Attention = Multi-Headed Self-Attention

Positional Encoding

Transformers – Conceptual Foundations

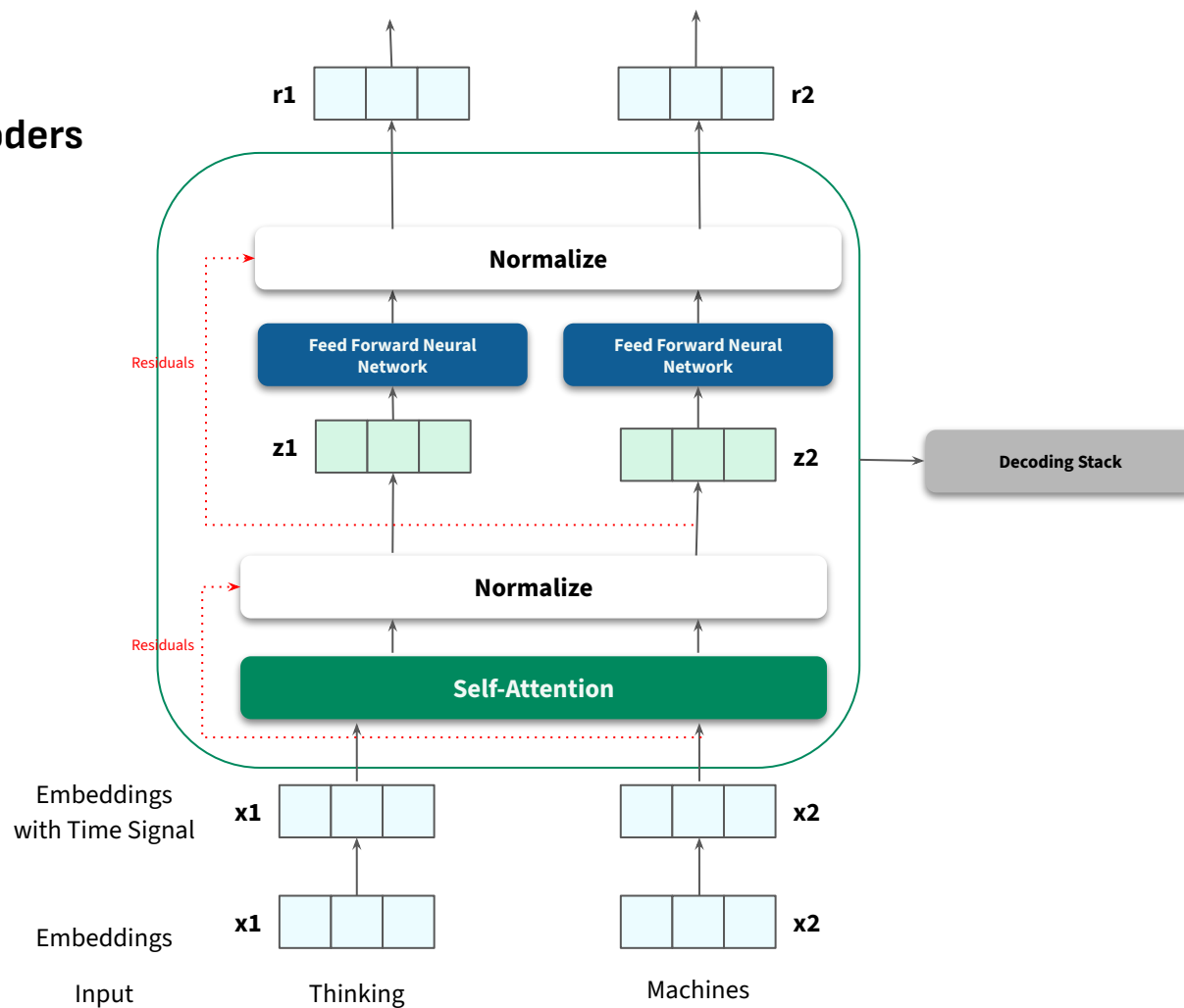
Final Encoding = Embedding + Position Encoding

Input	Thinking	Machines
Embeddings	x1 	x2 
Position	1	2
Positional Encoding	t1 	t2 
Final Encoding (Embedding with Time Signal)	x1 	x2 

Residuals & Layer Normalization

Transformers – Conceptual Foundations

Encoders

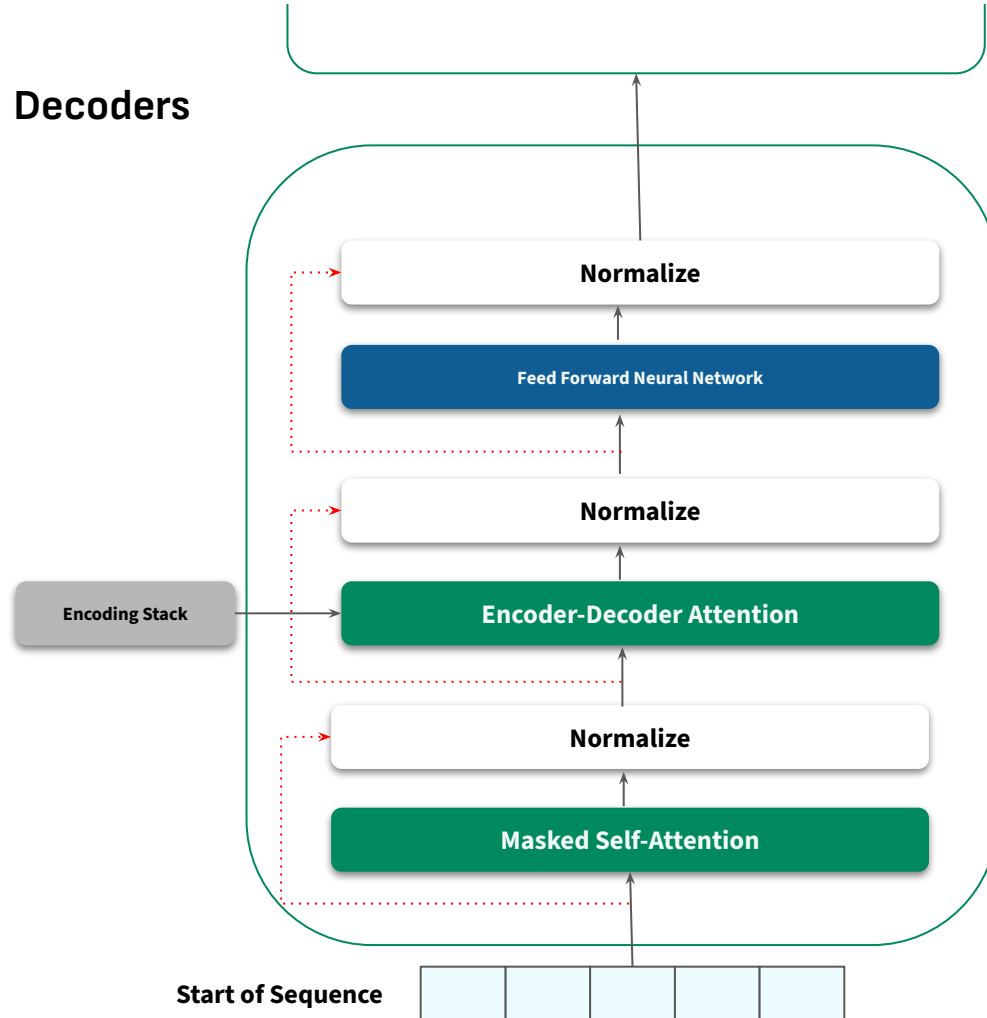


Decoders

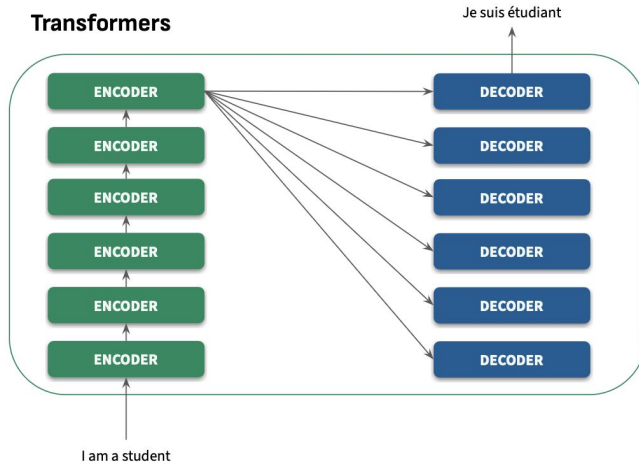
Transformers – Conceptual Foundations



Decoders



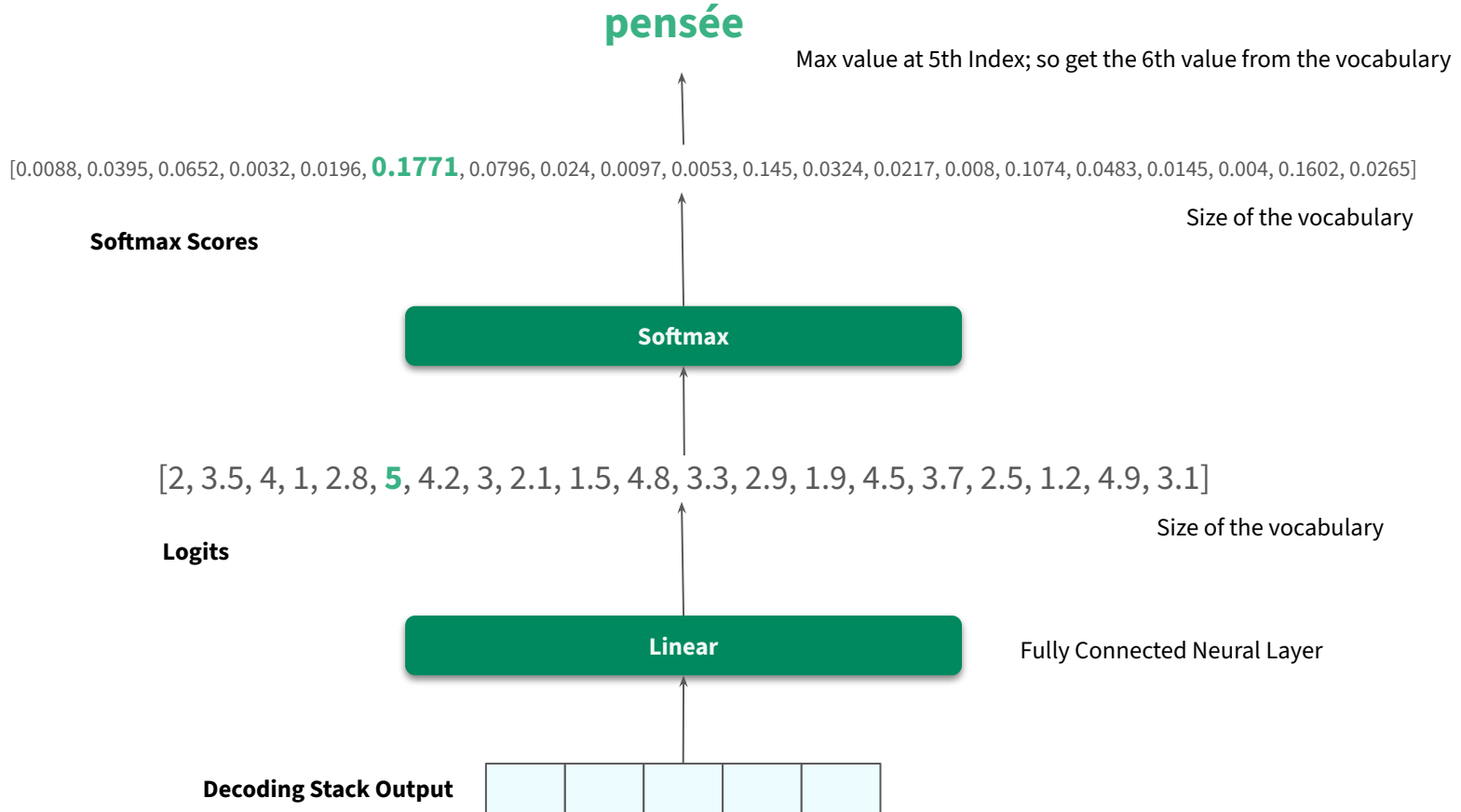
The Masked Self-Attention layer allows the decoder to weigh the importance of the different words in the output sequence generated so far.



Which part of the input is most important for me to look at right now to generate the next token?

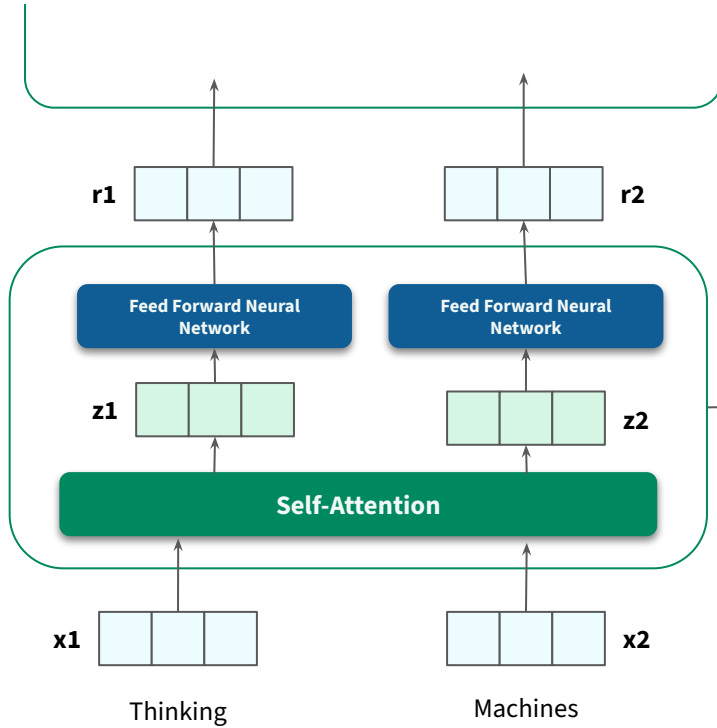
Linear and Softmax Layers

Transformers – Conceptual Foundations



Revisiting the High-Level Look

Transformers – Conceptual Foundations



Start of Sequence

